

Sandeep Gonnabattula

Software Engineer — Backend & AI Agent Systems

+91 7416303019 | gonnabattula19@gmail.com | linkedin.com/in/sandeep-go | github.com/sandeep6687

SUMMARY

Software Engineer with 18 months of production backend experience, including hands-on work building Python/FastAPI microservices for agentic AI workflow automation — LLM prompt-chaining, tool-calling orchestration, and pgvector-backed semantic retrieval to ground workflow decisions in real enterprise data. Independently delivered 50+ RESTful APIs in python across enterprise workflow and lead-management platforms, with deep experience in event-driven architecture (Kafka, Redis) and SQL Server/PostgreSQL optimization. Comfortable translating ambiguous requirements into scoped, shippable systems and iterating quickly using AI-assisted development workflows.

CORE COMPETENCIES

- Agentic AI Workflow Orchestration
- LLM Tool-Calling & Prompt Chaining
- Retrieval-Augmented Generation (pgvector)
- Backend API Development (C#/.NET, Python/FastAPI)
- Event-Driven Architecture (Kafka/Redis)
- SQL Server & PostgreSQL Optimization
- Microservices & System Design

TECHNICAL SKILLS

AI / LLM & Agent Systems: LLM integration (Gemini, Claude), prompt chaining, tool-calling workflow design, agentic task orchestration, embedding-based semantic retrieval (pgvector), context grounding for workflow decisions

Languages & Core CS: Python | Data Structures, Algorithms, OOP, System Design, Distributed Systems

Backend & APIs: FastAPI, Web API/RESTful, Microservices

Data & Messaging: SQL Server (query & index optimization), PostgreSQL, pgvector, Redis, Kafka, Azure Service Bus, Background Jobs

Cloud & DevOps: Azure (Functions/Services), Docker, Kubernetes

Tools & Testing: Git, Bitbucket, Postman, NUnit, Coverlet, Unit & Integration Testing

EXPERIENCE

Software Engineer | Zenoti

Jan 2025 – Jun 2026

- Built a workflow automation platform with workflow registration, triggers, conditions, multi-step execution, and distributed workers to automate complex business processes — the same orchestration and task-execution patterns central to agent-loop and plan/execute systems.
- Designed and developed microservices and event-driven architectures using Kafka, Redis, Docker, Kubernetes, and KEDA to support scalable, asynchronous processing of long-running workflows.
- Engineered production-grade REST APIs and backend services using python, PostgreSQL, and SQL Server for enterprise SaaS applications.
- Implemented Redis caching and database optimizations to reduce redundant database operations and improve backend performance and API responsiveness.
- Owned features end-to-end within Agile/Scrum delivery — requirement analysis, implementation, debugging, code review, testing, and production deployment.
- Developed secure and reliable backend systems with JWT authentication, API validation, exception handling, Swagger/OpenAPI, unit testing, and integration testing.

PROJECTS

Enterprise AI Workflow & Automation Engine | Python, FastAPI, C#, .NET Core, Kafka, Redis, PostgreSQL, pgvector, Azure Functions

- Architected a scalable event-driven workflow backend with FastAPI services orchestrating multi-step, tool-calling automation nodes, using Kafka for async processing and Redis for caching — supporting up to 40 automation steps/second.
- Engineered reusable AI-powered workflow nodes that chain LLM prompts with external REST API calls, using pgvector for embedding-based context retrieval to ground workflow decisions in real enterprise data.
- Designed the node execution model to support conditional branching and retries, improving reliability of long-running automation chains.

AI-Enhanced Lead Management System | ASP.NET Core, C#, Web API, EF Core, SQL Server, LLM Integration

- Developed scalable Web APIs with AI-based lead scoring and LLM-driven summary extraction to help sales teams prioritize high-intent leads.
- Optimized SQL Server execution plans to accelerate enterprise data retrieval across large lead datasets.

TalentPulse AI — Smart ATS Resume Optimizer & Mock Interview Platform | .NET 8 Web API, C#, EF Core, Gemini LLM

- Built an end-to-end AI resume-optimization backend using .NET 8 Web API and Gemini LLMs to parse resumes, run keyword-gap analysis against job descriptions, and auto-generate ATS-compliant resumes with PDF/DOCX export.
- Engineered an adaptive mock-interview API that dynamically generates technical questions based on a candidate's missing skills, with real-time evaluation and feedback scoring.

EDUCATION

VNR Vignana Jyothi Institute of Engineering and Technology — B.Tech, Computer Science (AI & ML), 8.02 CGPA | 2021 – 2025